1. How many images and classes?

For the **exact Kaggle version you linked**:

Dataset size: approximately **20,400 images/files**. (Kaggle)

The Kaggle dataset explorer shows:

cropped_images (998 directories)
20.4k files

However, **do not write "998 classes"** in your project. The 998 directories correspond to source/video organization, not 998 Real/Fake classes. (Kaggle)

For your **AI DeepGuard Detection** project, if you are using it as a binary detector, your practical labels should be:

- **REAL**
- **FAKE / DEEPFAKE**

But the exact Kaggle card does **not explicitly document a balanced REAL-vs-FAKE class count** for this particular 20.4K-image version. So don't claim 10,200 Real + 10,200 Fake unless your downloaded files/labels actually show that.

2. Dataset Size

There are two different numbers you should know for your viva:

Kaggle processed dataset

20.4K files / approximately 20,400 cropped face images
726.34 MB dataset size on Kaggle. (Kaggle)

Original FaceForensics research dataset

The original FaceForensics paper describes **more than 500,000 frames from 1,004 videos**. (arXiv)

So say:

"We use the Kaggle-preprocessed FaceForensics dataset containing approximately 20.4K cropped facial images. The original FaceForensics dataset contains over 500,000 frames from approximately 1,000 videos."

3. Class / Label Information

The **original FaceForensics** dataset contains:

- Original / pristine videos
- Manipulated videos
- FaceForensics manipulation primarily based on **Face2Face**

The later **FaceForensics++** dataset expanded the manipulation methods to:

1. DeepFakes
2. Face2Face
3. FaceSwap
4. NeuralTextures

and contains original sequences alongside manipulated sequences. (GitHub)

For your project, the classification can therefore be represented as:

Input Image

↓

FaceForensics Dataset

↓

REAL FAKE

↓ ↓

Original Manipulated

Important: The Kaggle page you gave specifically describes its content as **deepfake faces**, so don't automatically describe every image in that Kaggle copy as having a documented REAL/FAKE label unless your downloaded directory structure or metadata confirms it. (Kaggle)

4. Compression

The **original FaceForensics++ dataset** provides multiple compression levels:

Compression	Meaning
Raw / c0	Lossless / essentially original-quality
c23	H.264, moderate compression
c40	H.264, strong compression

The official documentation states that c23 is a visually high-quality compressed version and c40 represents stronger compression. (GitHub)

For the **specific Kaggle 20.4K cropped-image dataset**, however, the Kaggle description **does not specify whether the 150×150 images correspond to c0, c23, or c40**. Therefore write:

Compression: The original FaceForensics++ dataset provides c0, c23 and c40 compression variants. The specific Kaggle cropped-image version does not document its compression level.

5. Demographics

This is an important point for your synopsis.

The dataset documentation does **not provide detailed demographic labels** such as:

- Age
- Gender
- Ethnicity
- Skin tone
- Geographic demographic distribution

The original videos were collected from **YouTube**, with mostly frontal, trackable faces. (niessnerlab.org)

Therefore:

Demographics: Not explicitly provided in the Kaggle dataset metadata.

Do **not** claim that the dataset is demographically balanced.

6. Preprocessing

The Kaggle dataset has a clearly documented preprocessing pipeline:

Original Videos

↓

Extract Frames

↓

Every 25th Frame

↓

Face Detection / Extraction

↓

facenet_pytorch

↓

Crop Face

↓

Resize to 150 × 150

↓

Manual False-Positive Removal

↓

Final Dataset

This information comes directly from the Kaggle dataset description. (Kaggle)

For your project, you can describe the preprocessing as:

Frame Sampling → Face Extraction → Face Cropping → 150×150 Resizing → False-Positive Removal → Model Input

7. Noise Handling

The Kaggle dataset description **does not mention a dedicated denoising algorithm** such as:

- Gaussian filtering
- Median filtering
- Bilateral filtering
- Wiener filtering
- Non-local means

Instead, the documented cleaning step is:

Manual removal of false-positive face detections.

The original FaceForensics++ dataset deliberately provides different compression levels, which can be used to study robustness against compression-related artifacts.

So for your report:

Noise Handling: No explicit denoising operation is documented for the Kaggle version. Dataset cleaning primarily involved manual removal of incorrect face detections. Compression variants in the original FaceForensics++ dataset can additionally be used to evaluate robustness to compression artifacts.